



# STAGE 5.1-5.3 MATHEMATICS

Year 9 Assessment Task 1

March 29, 2012

## General instructions

- Working time – 50 minutes.
- Write your answers on your own A4, lined paper.
- Marks may be deducted for careless or poorly arranged work.
- Commence each new question on a new page.
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- Board approved calculators may be used.
- All necessary working should be shown in every question.
- Attempt **all** questions.
- At the conclusion of the examination, bundle the sheets used in the correct order within this paper and hand to examination supervisors.

## Class (please ✓)

- ☐ 9M1 – Mr Ireland/Mr Berry
- ☐ 9M2 – Mr Lam
- ☐ 9M3 – Ms Ziazaris/Mr Lowe
- ☐ 9M4 – Mr Lin/Mr Lucas
- ☐ 9M5 – Mr Weiss/Mr Berry

NAME: ..... # SHEETS USED: .....

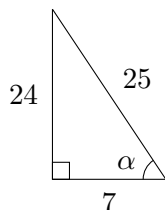
Marker's use only.

| QUESTION | 1  | 2  | 3 | 4  | 5  | Total | % |
|----------|----|----|---|----|----|-------|---|
| MARKS    | 13 | 16 | 5 | 13 | 11 | 58    |   |

**Question 1** (13 Marks) Commence a NEW page. **Marks**

(a) Evaluate  $\tan 70^\circ 46'$  correct to 2 decimal places. **2**

(b) Using the diagram, find these values as simplest fractions.

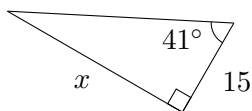


i.  $\tan \alpha$ . **1**

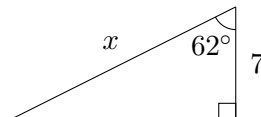
ii.  $2 \sin \alpha \cos \alpha$ . **2**

(c) Find the value of the pronumeral, correct to 1 decimal place/nearest minute. **8**

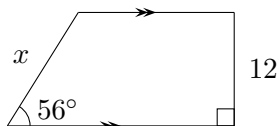
i.



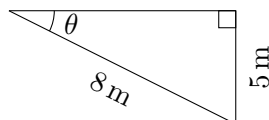
ii.



iii.



iv.



**Question 2** (16 Marks) Commence a NEW page. **Marks**

(a) Expand and write the following expressions in simplest form.

i.  $2(x^2 + 7t)$  **1**

ii.  $(4a + 1)(a + 3)$  **2**

iii.  $(5x - 2)^2$  **2**

(b) Factorise fully:

i.  $7x^2 - 49x$  **1**

ii.  $8xy + 2y + 12x + 3$  **2**

iii.  $x^2 - 49$  **2**

iv.  $x^2 - 5x - 6$  **2**

v.  $8x^2 - 10x - 3$  **2**

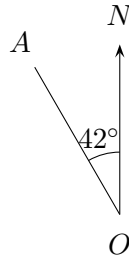
vi.  $8x^3 + 125$  **2**

**Question 3** (5 Marks)

Commence a NEW page.

**Marks**

- (a) Solve for  $x$ :  $\sin 4x = \cos (6x + 50)$ . **2**
- (b) Find the true (three figure) bearing of  $A$  from  $O$ . **1**



- (c) A ship sails on a bearing of  $156^\circ\text{T}$  until it is 45 km south of its starting point. How far east is it, correct to 2 decimal places? **2**

**Question 4** (13 Marks)

Commence a NEW page.

**Marks**

- (a) Simplify fully:
- $\frac{5a-7}{2a+4} \div \frac{10a-14}{12}$  **3**
  - $\frac{x^2+3x-4}{2x-4} \times \frac{6x-12}{x-1}$  **3**
- (b) Express the following as a single fraction in simplest form.
- $\frac{3}{2x+1} + \frac{1}{4-x}$  **3**
  - $\frac{3}{x^2-x-6} + \frac{4}{x^2-4x+3}$  **4**

**Examination continues overleaf...**

**Question 5** (11 Marks) Commence a NEW page. **Marks**

- (a) Expand and fully simplify **3**

$$(a - 5)^2 + (a - 4)^2 + \cdots + a^2 + \cdots + (a + 4)^2 + (a + 5)^2$$

- (b) Fully factorise  $4x^4 - 37x^2 + 9$  (Hint: Let  $m = x^2$ ) **3**

- (c) A passenger on a small cruise ship observed, whilst passing under the Harbour Bridge, that the angle of elevation to the pylons are  $47^\circ$  and  $28^\circ$ . It is known that the distance between the pylons is 503 m. (Assume the pylons are equal in height and elevation.)



- i. By drawing an appropriate diagram, labelling sides and assigning variables, show that the height  $h$  of the pylons (in metres) is given by **3**

$$h = \frac{503 \tan 47^\circ \tan 28^\circ}{\tan 47^\circ + \tan 28^\circ}$$

- ii. Hence or otherwise, find the height of the pylon in centimetres, using scientific notation correct to 4 significant figures. **2**

🐼 End of paper. 🐼

**Suggested solutions****Question 1** (13 marks)

(a) (2 marks)

✓ [1] for magnitude.

✓ [1] for rounding.

$$\tan 70^\circ 46' = 2.87 \text{ (2 d.p.)}$$

(b) i. (1 mark)

$$\tan \alpha = \frac{24}{7}$$

ii. (2 marks)

✓ [-1] for each error.

$$\begin{aligned} 2 \sin \alpha \cos \alpha &= 2 \times \frac{24}{25} \times \frac{7}{25} \\ &= \frac{336}{625} \end{aligned}$$

(c) ✓ [2] for all correct.

✓ [-1] for each error. No penalty for rounding except for extreme cases.

i.

$$\begin{aligned} \frac{x}{15} &= \tan 41^\circ \\ x &= 15 \tan 41^\circ = 13.039 \dots \end{aligned}$$

ii.

$$\begin{aligned} \frac{7}{x} &= \cos 62^\circ \\ x &= \frac{7}{\cos 62^\circ} = 14.9 \dots \end{aligned}$$

iii.

$$\begin{aligned} \frac{12}{x} &= \sin 56^\circ \\ x &= \frac{12}{\sin 56^\circ} = 14.47 \dots \end{aligned}$$

iv.

$$\begin{aligned} \sin \theta &= \frac{5}{8} \\ \theta &= 38^\circ 41' \end{aligned}$$

**Question 2** (16 marks)

(a) i. (1 mark)

$$2(x^2 + 7t) = 2x^2 + 14t$$

ii. (2 marks)

✓ [-1] for each error.

$$(4a + 1)(a + 3) = 4a^2 + 13a + 3$$

iii. (2 marks)

✓ [-1] for each error.

✓ [0] if  $25x^2 + 4$  or  $25x^2 - 4$  is written down.

$$(5x - 2)^2 = 25x^2 - 20x + 4$$

(b) i. (1 mark)

$$7x^2 - 49x = 7x(x - 7)$$

ii. (2 marks)

✓ [1] for  $2y(4x + 1) + 3(4x + 1)$ .

✓ [1] for fully factorising.

$$\begin{aligned} 8xy + 2y + 12x + 3 &= 2y(4x + 1) + 3(4x + 1) \\ &= (2y + 3)(4x + 1) \end{aligned}$$

iii. (2 marks)

✓ [-1] for each error.

$$x^2 - 49 = (x - 7)(x + 7)$$

iv. (2 marks)

✓ [-1] for each error.

$$x^2 - 5x - 6 = (x - 6)(x + 1)$$

v. (2 marks)

✓ [-1] for each error.

$$8x^2 - 10x - 3 = (4x + 1)(2x - 3)$$

vi. (2 marks)

✓ [-1] for each error.

$$8x^3 + 125 = (2x + 5)(4x^2 - 10x + 25)$$



**Question 5** (11 marks)

(a) (3 marks)

- ✓ [1] expanding some terms correctly.
- ✓ [1] notice pattern in expansions.
- ✓ [1] final answer.

$$(a - 5)^2 = a^2 - 10a + 25$$

$$(a + 5)^2 = a^2 + 10a + 25$$

$$(a - 4)^2 = a^2 - 8a + 16$$

$$(a + 4)^2 = a^2 + 8a + 16$$

$$\vdots$$
**Observations**

- there will be eleven  $a^2$  terms (including the single  $(a + 0)^2$  expansion)
- all  $a$  terms are offset;
- Constant terms produces double of sum of squares –  $2(1^2 + 2^2 + \dots + 5^2)$

$$\begin{aligned} \therefore (a - 5)^2 + \dots + a^2 + \dots + (a + 5)^2 \\ = 11a^2 + 110 \end{aligned}$$

(b) (3 marks)

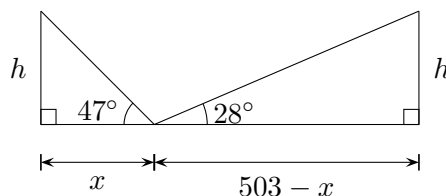
- ✓ [1] for making correct substitution.
- ✓ [1] for factorising in  $m$
- ✓ [1] for fully factorising in  $x$ .

Letting  $m = x^2$ ,

$$\begin{aligned} 4x^4 - 37x^2 + 9 \\ = 4m^2 - 37m + 9 \\ = (4m - 1)(m - 9) \\ = (4x^2 - 1)(x^2 - 9) \\ = (2x - 1)(2x + 1)(x - 3)(x + 3) \end{aligned}$$

(c) i. (3 marks)

- ✓ [1] for correct diagram
- ✓ [1] for two equations in  $x$ , and  $503 - x$
- ✓ [1] for final answer.



Let  $x$  be the distance from the souther pylon to the ship (could be from the northern pylon, but this is not important as the solution is identical)

$$\begin{aligned} \frac{h}{x} &= \tan 47^\circ \\ \therefore x &= \frac{h}{\tan 47^\circ} \end{aligned} \quad (5.1)$$

In the ther triangle,

$$\begin{aligned} \frac{h}{503 - x} &= \tan 28^\circ \\ \therefore 503 - x &= \frac{h}{\tan 28^\circ} \\ x &= 503 - \frac{h}{\tan 28^\circ} \end{aligned} \quad (5.2)$$

Equating  $x$  in (5.1) and (5.2)

$$\begin{aligned} 503 - \frac{h}{\tan 28^\circ} &= \frac{h}{\tan 47^\circ} \\ 503 &= \frac{h}{\tan 47^\circ} + \frac{h}{\tan 28^\circ} \\ &= h \left( \frac{1}{\tan 47^\circ} + \frac{1}{\tan 28^\circ} \right) \\ &= h \left( \frac{\tan 28^\circ + \tan 47^\circ}{\tan 28^\circ \tan 47^\circ} \right) \end{aligned}$$

Dividing both sides by  $\frac{\tan 28^\circ + \tan 47^\circ}{\tan 28^\circ \tan 47^\circ}$ ,

$$\therefore h = \frac{503 \tan 28^\circ \tan 47^\circ}{\tan 28^\circ + \tan 47^\circ}$$

ii. (2 marks)

- ✓ [1] for correct number of significant figures.
- ✓ [1] for correct scientific notation.
- ✓ [-1] for metres (required to give cm)

$$h = 178.79 \text{ m} = 1.788 \times 10^4 \text{ cm}$$